Higher-order galaxy bias for DESI thin-bin angular clustering

A theoretical study for the lssty1xdesi project

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Abstract

The lssty1xdesi forecast combines LSST Y1 3x2pt with DESI angular galaxy clustering C_ℓ^{gg} in thin redshift bins, computed without the Limber approximation. Its likelihood models galaxy clustering with linear bias times the halofit matter power spectrum, and the draft paper extends that to a one-loop Eulerian bias expansion. This note collects the physics that any higher-order bias implementation must respect: the one-loop galaxy power spectrum in real and redshift space, the Eulerian and Lagrangian bias bases with every conversion factor, the DESI DR1 full-shape parameterization and priors, the projection of each term to thin-bin angular spectra, and the measured size of each term at the project's scale cuts. Two conclusions carry the most weight. First, the draft paper's bias table holds the DESI DR1 "physical" sampling combinations $(1+b_1^{\rm L})\sigma_8(z_{\rm eff})$, $b_2^{\rm L}\sigma_8^2$ and $b_s^{\rm L}\sigma_8^2$, inserted into the Eulerian calculator without removing the σ_8 powers, without rotating from the Lagrangian to the Eulerian basis, and without the factor of two of the CCL normalization. The linear-bias column confirms this reading for four of the five chunks; the BGS entry does not match the project fiducial and needs a check against the DESI chains. Under this reading the tidal term enters the galaxy power spectrum with the opposite sign and several times the magnitude of the co-evolution expectation, which moves the loop correction by several percent of P_{gg} at $k \simeq 0.1\,h\,\mathrm{Mpc}^{-1}$, the size of the whole effect. Second, at the project cut $k_{\rm max} = 0.075\,h\,\mathrm{Mpc}^{-1}$ the one-loop terms are a 1–3% correction that a minimal model (co-evolution-centered b_2 and b_{s^2}, one counterterm, one stochastic amplitude per tracer chunk) covers; at the shipped mask near $0.18\,h\,\mathrm{Mpc}^{-1}$ the corrections reach 10–20% and require the full one-loop parameter set per chunk, including the redshift-space pieces this project has not yet modeled.

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1 Scope and context

The lssty1xdesi project forecasts DESI spectroscopic galaxy clustering as angular power spectra C_ℓ^{gg} in 76 thin bins of equal comoving width (about 62 Mpc), combined with LSST Y1 3x2pt. The thin bins resolve radial Fourier modes up to $k_\parallel \simeq \pi/\Delta\chi \simeq 0.05\,\mathrm{Mpc}^{-1} \simeq 0.075\,h\,\mathrm{Mpc}^{-1}$, where $\Delta\chi$ is the comoving bin width, so the angular analysis recovers most of the three-dimensional information of a power spectrum analysis at the same scale cut [11]. The shipped likelihood models the galaxy power as $P_{gg} = b_1^2(z)\,P_{\rm mm}(k,z)$ with $P_{\rm mm}$ the halofit matter power spectrum, one bias amplitude per tracer chunk evolving as $b_1(z) = B_g/D(z)$, where $D(z)$ is the linear growth factor normalized to $D(0) = 1$ and B_g is the chunk amplitude. The $1/D$ shape keeps the clustering amplitude $b_1(z)D(z)$ of a chunk constant; it is the constant-clustering-amplitude shape, not the evolution law of a conserved galaxy population.

An earlier review of the forecast (the project reference `references/fom_and_bias_review.md`) concluded that this model is too rigid for the shipped scale cuts: linear bias is accurate only to $k_{\rm max} \simeq 0.08$–$0.12\,h\,\mathrm{Mpc}^{-1}$, while the shipped masks reach $k \simeq 0.18\,h\,\mathrm{Mpc}^{-1}$. The missing terms are the second-order and tidal bias loops, the k^2 counterterms, and the stochastic terms. This note is the theoretical study that the higher-order bias implementation will build on. It answers six questions: the structure of the one-loop galaxy power spectrum and its bias bases (Section 2,

derivations in Appendix A); the DESI DR1 full-shape model and priors (Section 3); the origin of the draft paper's bias values and the tidal-bias verdict (Section 4); the projection of each term to thin-bin angular spectra (Section 5); the measured size of every term (Section 6); and the recommended model (Sections 8 and 9). Section 10 reports the companion implementation study: the CosmoLike inventory, the comparison with the draft's data vector, and the ordered plan.

Conventions. k is the comoving wavenumber, quoted in h/Mpc unless a unit is written; z is redshift; μ is the cosine between the wavevector and the line of sight; $f = d\ln D/d\ln a$ is the linear growth rate; $\sigma_8(z)$ is the rms linear density fluctuation in spheres of $8\,h^{-1}$Mpc; A_s is the primordial scalar amplitude; $\bar{n}$ is the tracer number density. The matter overdensity is δ, the galaxy overdensity δ_g, and $s_{ij} = (\partial_i\partial_j/\partial^2 - \delta_{ij}/3)\,\delta$ is the tidal field with $s^2 = s_{ij}s_{ij}$.

2 The one-loop galaxy power spectrum

2.1 Real space, Eulerian basis

At one loop, every rotationally invariant local gravitational observable up to third order in the linear density enters the galaxy density. After renormalization, the McDonald–Roy basis [1, 2] carries four deterministic bias parameters plus a derivative bias and a noise field:

$$\delta_g = b_1\,\delta + \frac{b_2}{2}\left(\delta^2 - \langle\delta^2\rangle\right) + \frac{b_{s^2}}{2}\left(s^2 - \langle s^2\rangle\right) + \frac{b_{3\mathrm{nl}}}{2}\,\psi + b_{\nabla^2}\,\nabla^2\delta + \epsilon, \tag{1}$$

where ψ is the one independent third-order operator that survives at one loop (Appendix A writes it out) and ϵ is the stochastic field uncorrelated with the long modes. This is the convention of FAST-PT and of CCL's `EulerianPTCalculator`, verified in this study against the CCL source (`pyccl/nl_pt/ept.py`): the code documents the expansion of Eq. (1) and assembles the auto spectrum as

$$\begin{aligned} P_{gg}(k) = {} & b_1^2\,P_{\mathrm{mm}}(k) + b_1 b_2\,P_{\delta_1\delta_2}(k) + \tfrac{1}{4}b_2^2\,P_{\delta_2\delta_2}(k) + b_1 b_{s^2}\,P_{\delta_1 s_2}(k) \\ & + \tfrac{1}{2}b_2 b_{s^2}\,P_{\delta_2 s_2}(k) + \tfrac{1}{4}b_{s^2}^2\,P_{s_2 s_2}(k) + b_1 b_{3\mathrm{nl}}\,\sigma_3^2(k)\,P_{\mathrm{lin}}(k) + b_1 b_{k^2}k^2 P(k), \end{aligned} \tag{2}$$

plus stochastic terms discussed below. The six quadratic templates P_{XY} are the mode-coupling integrals of two linear spectra over the second-order kernels (Appendix B); $\sigma_3^2(k)$ is the third-order filter of [1, 2]; and b_{k^2} is CCL's name for the derivative-bias coefficient, whose term is degenerate at this order with the effective-field-theory counterterm $\alpha_0 k^2 P_{\mathrm{lin}}$. The draft paper's six-term expression is exactly the first six terms of Eq. (2), as the earlier review already established; the draft then adds $\alpha_0 k^2 P_{\mathrm{lin}}$ and sets $b_{3\mathrm{nl}} = 0$.

Three templates do not vanish at low k: $P_{\delta_2\delta_2} \to 2\sigma_4$, $P_{\delta_2 s_2} \to \frac{4}{3}\sigma_4$ and $P_{s_2 s_2} \to \frac{8}{9}\sigma_4$, with σ_4 a constant built from the linear spectrum (Appendix B). These constants are white noise, indistinguishable from shot noise, and CCL subtracts them only when its `sub_lowk` option is on (the default is off). Whichever setting the pipeline uses must be stated, because with `sub_lowk` off and no free shot-noise parameter the constants enter the data vector as unmodeled power: at the co-evolution biases of the BGS chunk the three constants sum to about 920 $(h^{-1}\mathrm{Mpc})^3$ at $z = 0.3$ (the $\frac{1}{4}b_2^2$ piece alone is 517), five times that chunk's Poisson power $1/\bar{n} \simeq 170$ $(h^{-1}\mathrm{Mpc})^3$. CosmoLike's galaxy-clustering integrand subtracts exactly these constants (Section 10), the `sub_lowk`-on convention, so a CCL vector at the CCL default and a CosmoLike vector differ by this white noise until the two settings are aligned.

The hybrid choice of Eq. (2) is that the b_1^2 term carries the full nonlinear matter power (halofit here) while the loop templates stay at one-loop standard perturbation theory. Nicola et al. [9] validated this hybrid against simulations for an LSST-like projected 3x2pt analysis: with the nonlinear $P_{\rm mm}$ in the b_1^2 term the Eulerian model recovers unbiased cosmology to at least $k_{\rm max} = 0.4\,{\rm Mpc}^{-1}$, while the fully perturbative $P_{\rm mm}$ fails beyond about $0.2\,{\rm Mpc}^{-1}$. The same work finds significant non-Poisson stochasticity in every galaxy sample it fits, a point Section 6 returns to.

2.2 The Lagrangian basis and the exact conversions

The velocileptors code, the DESI DR1 baseline, expands the galaxy field in the initial (Lagrangian) coordinates [5, 6]:

$$F[\delta_0(\mathbf{q})] = 1 + b_1^{\rm L}\,\delta_0 + \frac{b_2^{\rm L}}{2}\left(\delta_0^2 - \langle\delta_0^2\rangle\right) + b_s^{\rm L}\left(s_0^2 - \langle s_0^2\rangle\right) + b_3^{\rm L}\,O_3(\mathbf{q}), \tag{3}$$

where δ_0 and s_0 are the linear density and tidal fields at the initial time and O_3 is the one independent cubic operator. Note the normalization: the quadratic term carries a $1/2$, the tidal term does not. Chen, Vlah and White [5] give the rotation to their Eulerian parameters c_n, defined by $\delta_g = c_1\delta + \frac{c_2}{2}\delta^2 + c_s s^2 + \ldots$ (again no $1/2$ on the tidal term):

$$c_1 = 1 + b_1^{\rm L}, \qquad c_2 = b_2^{\rm L} + \frac{8}{21}\,b_1^{\rm L}, \qquad c_s = b_s^{\rm L} - \frac{2}{7}\,b_1^{\rm L}, \qquad c_3 = b_3^{\rm L} + a\,b_1^{\rm L}, \tag{4}$$

with a a constant fixed by the choice of third-order operator. Appendix A derives the $8/21$ and $-2/7$ from the second-order co-evolution solution. Mapping the c_n onto the CCL convention of Eq. (1) costs one more step, the factor of two between $c_s s^2$ and $(b_{s^2}/2)s^2$:

$$\boxed{b_1 = 1 + b_1^{\rm L}, \qquad b_2 = b_2^{\rm L} + \frac{8}{21}\,b_1^{\rm L}, \qquad b_{s^2} = 2\,b_s^{\rm L} - \frac{4}{7}\,b_1^{\rm L}, \qquad b_{\rm 3nl} = \frac{32}{315}\,b_1^{\rm L} + 2\,b_3^{\rm L}.} \tag{5}$$

The $b_{\rm 3nl}$ entry uses the third-order co-evolution result of Saito et al. [2] for the dynamically generated part; the $2b_3^{\rm L}$ normalization depends on the choice of O_3 and is irrelevant here because every analysis in play sets $b_3^{\rm L} = 0$. A third common basis writes the tidal operator as the Galileon $\mathcal{G}_2 = s^2 - \frac{2}{3}\delta^2$ with coefficient $b_{\mathcal{G}_2}$ (written γ_2 by [10]); then $b_{\mathcal{G}_2} = c_s = b_{s^2}/2$ and the quadratic coefficient shifts by $+\frac{4}{3}b_{\mathcal{G}_2}$ (Appendix A).

2.3 Co-evolution and the $b_n(b_1)$ calibrations

If galaxies form with purely local Lagrangian bias ($b_s^{\rm L} = b_3^{\rm L} = 0$) and are then advected by gravity, the Eulerian tidal and third-order biases are generated by the dynamics alone. Setting $b_s^{\rm L} = 0$ in Eq. (5) gives the co-evolution relations [2]

$$b_{s^2} = -\frac{4}{7}\,(b_1 - 1)\,, \qquad b_{\rm 3nl} = \frac{32}{315}\,(b_1 - 1)\,. \tag{6}$$

Both are negative-slope in b_1 for the tidal bias: every tracer with $b_1 > 1$ has $b_{s^2} < 0$ in this convention. Simulations support the relations at the 10–30% level; the excursion-set calibration used by [10], $b_{\mathcal{G}_2} = 0.524 - 0.547\,b_1 + 0.046\,b_1^2$, gives a $b_{s^2} = 2b_{\mathcal{G}_2}$ about 30% more negative than

chunk	$z_{\rm eff}$	$b_1 = B_g/D$	b_2 (Lazeyras)	b_{s^2} (co-ev.)	$b_{\rm 3nl}$ (co-ev.)	b_{s^2} at chunk edges
BGS	0.295	1.56	-0.64	-0.32	$+0.06$	-0.24 to -0.37
LRG1	0.510	2.21	$+0.30$	-0.69	$+0.12$	-0.62 to -0.75
LRG2	0.706	2.43	$+0.81$	-0.82	$+0.15$	-0.75 to -0.88
LRG3	0.919	2.68	$+1.50$	-0.96	$+0.17$	-0.88 to -1.09
ELG1	0.955	1.35	-0.77	-0.20	$+0.04$	-0.15 to -0.25
ELG2	1.317	1.57	-0.64	-0.32	$+0.06$	-0.25 to -0.42

Table 1: Higher-order bias expectations per DESI chunk in the CCL convention of Eq. (1), from the project's fiducial linear biases $b_1 = B_g/D(z)$ at the chunk effective redshifts, the Lazeyras et al. quadratic fit of Eq. (7), and the co-evolution relations of Eq. (6). The growth factor is computed at the project fiducial cosmology ($\Omega_m = 0.3$, $h = 0.6732$). The $z_{\rm eff}$ values are the DESI DR1 effective redshifts; ELG1, which the DESI DR1 full-shape analysis excluded, uses its chunk's central redshift.

Eq. (6) at $b_1 \simeq 2.2$. For the quadratic bias the separate-universe fit of Lazeyras et al. [3], valid for $1 \lesssim b_1 \lesssim 10$ and redshift independent, reads

$$b_2(b_1) = 0.412 - 2.143\, b_1 + 0.929\, b_1^2 + 0.008\, b_1^3. \tag{7}$$

Table 1 evaluates these expectations for the six DESI chunks of this project, with $b_1 = B_g/D(z_{\rm eff})$ and the project amplitudes $B_g = 1.34$ (BGS), 1.70 (LRG) and 0.84 (ELG). The last two columns show how much the co-evolution b_{s^2} moves across a chunk when b_1 follows the $B_g/D(z)$ shape; this drift is the subject of Section 8.

2.4 Redshift space

In redshift space the one-loop model gains no new bias parameters but several new terms. The DESI DR1 methodology paper [7] writes every full-shape model in the generic form

$$P_{s,g}(k,\mu) = P^{\rm PT}_{s,g}(k,\mu) + \left(\alpha_0 + \alpha_2\mu^2 + \alpha_4\mu^4\right) k^2 P_{\rm lin}(k,\mu) + \left({\rm SN}_0 + {\rm SN}_2\, k^2\mu^2 + {\rm SN}_4\, k^4\mu^4\right). \tag{8}$$

$P^{\rm PT}_{s,g}$ is the one-loop redshift-space spectrum of biased tracers. Its tree-level limit is the Kaiser spectrum $(b_1 + f\mu^2)^2\, P_{\rm lin}$; at one loop it couples every bias operator of Eq. (3) to the velocity field, producing μ-dependent loop terms with the same four bias coefficients (the complete velocileptors expression is the resummed one-loop redshift-space power spectrum in the theory section of [6], built from the displacement-field correlators of [5]). The three counterterms $\alpha_0, \alpha_2, \alpha_4$ absorb the small-scale sensitivity of the loop integrals and the leading velocity dispersion; velocileptors normalizes them as fractional corrections to the linear multipoles, so that $\alpha_n = 12.5\, h^{-2}{\rm Mpc}^2$ is a 50% correction to the n-th moment at $k = 0.2\, h\,{\rm Mpc}^{-1}$ [6]. The three stochastic terms descend from the noise field: ${\rm SN}_0$ is the deviation of the constant noise from the Poisson value already subtracted from the estimator, ${\rm SN}_2$ is the velocity-weighted noise (the leading Fingers-of-God stochastic term), and ${\rm SN}_4$ its next moment.

The structural summary: a one-loop redshift-space analysis carries per tracer sample four deterministic biases $(b_1^{\rm L}, b_2^{\rm L}, b_s^{\rm L}, b_3^{\rm L})$, three counterterms, and three stochastic amplitudes; a real-space or projected analysis keeps the same biases but collapses the counterterms toward α_0 and the stochastic tower toward ${\rm SN}_0$, with the μ-dependent members reappearing in exactly the way Section 5 describes.

3 The DESI DR1 full-shape model and priors

The DESI DR1 full-shape analysis [7, 8] fits the power spectrum monopole and quadrupole of six tracer bins (BGS at $z_{\rm eff} = 0.295$, LRG1–3 at 0.510, 0.706, 0.919, ELG2 at 1.317, QSO at 1.491) over $0.02 < k < 0.2\,h\,\mathrm{Mpc}^{-1}$. The ELG1 bin ($0.8 < z < 1.1$) failed the fibre-collision systematics tests and is excluded from the full-shape analysis [7]. Four theory codes were validated against each other and simulations; the baseline is the Eulerian mode of velocileptors, with Folps and PyBird as cross-checks and a Gaussian-streaming configuration-space model as a fourth [7]. The parameter choices that matter here:

- **Basis and sampling.** Even in the Eulerian mode, DESI samples the Lagrangian parameters of Eq. (3), reparameterized as the "physical" combinations

$$\left(1 + b_1^{\rm L}\right)\sigma_8(z_{\rm eff}), \qquad b_2^{\rm L}\,\sigma_8^2(z_{\rm eff}), \qquad b_s^{\rm L}\,\sigma_8^2(z_{\rm eff}), \tag{9}$$

  because the data constrain these products directly and the reparameterization reduces projection effects on σ_8 [7, 8]. Note that $1 + b_1^{\rm L}$ is the Eulerian linear bias, so the first combination is $b_1\sigma_8(z_{\rm eff})$.

- **Priors.** Uniform $[0, 3]$ on $(1+b_1^{\rm L})\sigma_8$; Gaussian with mean zero and width 5 on $b_2^{\rm L}\sigma_8^2$ and on $b_s^{\rm L}\sigma_8^2$; Gaussian with mean zero and width $12.5\,h^{-2}\mathrm{Mpc}^2$ on α_0 and α_2 (the 50%-correction-at-$k_{\rm max}$ rule); Gaussian widths of 2 Poisson units on SN_0 and of 5 units of $f_{\rm sat}\sigma_v^2/\bar{n}$ on SN_2, where $f_{\rm sat}$ is the satellite fraction and σ_v a characteristic velocity dispersion [7, 8].

- **Fixed parameters.** $b_3^{\rm L} = 0$, because the cubic bias is expected to be small for intermediate-mass halos and is degenerate with the counterterms [7, 6]; α_4 and SN_4 drop with the hexadecapole, which DESI DR1 does not fit. The tidal bias is deliberately left free rather than tied to co-evolution: tying it would add cosmological information that DESI chose not to claim [7].

- **Budget.** Seven nuisance parameters per bin ($b_1^{\rm L}, b_2^{\rm L}, b_s^{\rm L}, \alpha_0, \alpha_2, \mathrm{SN}_0, \mathrm{SN}_2$), 42 in total, with the counterterms and stochastic terms marginalized analytically [8].

Semenaite et al. [10] fit the same DESI DR1 tracers in configuration space jointly with weak lensing, with one structural difference relevant to this project: both tidal biases are fixed to calibrated relations in b_1 (the excursion-set $\gamma_2(b_1)$ and the conserved-evolution $\gamma_{21}(b_1, \gamma_2)$), leaving b_1 and b_2 free per bin with flat priors, plus counterterms and a virial damping parameter. Their validation on mocks shows that fixing the tidal sector to these relations recovers unbiased cosmology for DESI-like tracers, which is direct evidence for the co-evolution-anchored minimal model this note recommends.

4 The draft paper's bias table: the tidal-bias verdict

The draft paper's bias table (its Table of "bias parameters per DESI redshift bin") lists, per chunk, values labeled $b_1^{\rm E}$, $b_2^{\rm E}$, $b_s^{\rm E}$, attributed to the DESI DR1 full-shape analysis. The earlier review flagged the $b_s^{\rm E}$ column (positive, 0.6–1.4) as incompatible with co-evolution, which predicts b_{s^2} between -0.3 and -1.0 for these tracers (Table 1), and suspected a Lagrangian-to-Eulerian conversion error without establishing it. The provenance can now be established from the numbers themselves.

chunk	$z_{\rm eff}$	draft "$b_1^{\rm E}$"	$\sigma_8(z_{\rm eff})$	draft value / $\sigma_8(z_{\rm eff})$	project $b_1 = B_g/D$
BGS	0.295	1.37500	0.724	1.90	1.56
LRG1	0.510	1.44800	0.649	2.23	2.21
LRG2	0.706	1.40300	0.590	2.38	2.43
LRG3	0.919	1.38800	0.535	2.60	2.68
ELG2	1.317	0.70400	0.453	1.56	1.56

Table 2: The draft table's linear-bias column read as the DESI physical combination $b_1\sigma_8(z_{\rm eff})$. Dividing each entry by σ_8 at the DESI DR1 effective redshift of that chunk (computed at the project fiducial cosmology, $\sigma_8(0) = 0.843$) returns an Eulerian linear bias that matches the project's own fiducial $B_g/D(z_{\rm eff})$ to 1–5% for LRG1–3 and ELG2. The BGS row differs by 27%; the text discusses it.

Three facts identify the table as the DESI physical-basis combinations of Eq. (9) inserted raw:

1. **The linear column divided by $\sigma_8(z_{\rm eff})$ lands on the known biases for four of the five chunks.** Table 2 shows the check: for LRG1–3 and ELG2 the implied Eulerian bias agrees with the project's fiducial $B_g/D(z_{\rm eff})$ to 1–5%. An independent check at a Planck-like cosmology ($\Omega_m = 0.3138$, $\sigma_8 = 0.8076$, run by the implementation study) finds the same pattern: ratios of the implied bias to $B_g/D(z_{\rm eff})$ of 1.055, 1.022, 1.011 and 1.038 for LRG1–3 and ELG2. Read directly as Eulerian biases instead, the column is unphysical: it would make the three LRG chunks nearly unbiased ($b_1 \simeq 1.4$ where LRGs sit near 2.2–2.7) and the ELG2 chunk anti-biased ($b_1 = 0.70$) at $z = 1.3$. The BGS row is the exception: it implies $b_1 \simeq 1.90$ against the project's 1.56, a ratio of 1.27 in both checks. Whether that reflects the project's BGS fiducial $B_g = 1.34$, which comes from the source forecast's configuration rather than from a DESI DR1 fit, or a different origin of that row, is not established here. The direct test is to read the BGS entry of the public DESI DR1 full-shape chains: if the maximum-posterior $(1 + b_1^{\rm L})\sigma_8$ is 1.375, the reading covers BGS as well and the project's BGS amplitude is the quantity to revisit.

2. **The empty ELG1 row.** DESI DR1 excluded ELG1 from the full-shape analysis (Section 3); a table transcribed from the DESI full-shape products has nothing to put there, and the draft's row is indeed empty.

3. **The magnitudes of the other two columns.** Read as $b_2^{\rm L}\sigma_8^2$ and $b_s^{\rm L}\sigma_8^2$, the draft's second and third columns (-1.5 to $+0.3$ and $+0.6$ to $+1.4$) sit comfortably inside the DESI Gaussian priors of width 5 on exactly those combinations. Read as CCL Eulerian parameters they contradict both co-evolution and the DESI prior centering.

Verdict. The draft's bias table holds the DESI DR1 sampling combinations $(1+b_1^{\rm L})\sigma_8(z_{\rm eff})$, $b_2^{\rm L}\sigma_8^2$ and $b_s^{\rm L}\sigma_8^2$, used as CCL's Eulerian (b_1, b_2, b_{s^2}) with three transformations still to apply: removing the σ_8 powers, rotating the Lagrangian parameters to the Eulerian basis via Eq. (5), and the factor of two between the velocileptors and CCL tidal normalizations. The reading is established by the linear column for LRG1–3 and ELG2 and extends to the other two columns by the table's construction; the BGS linear entry awaits the chain check above. The sign puzzle dissolves: the column was never an Eulerian tidal bias. The corrected recipe, if DESI values are

wanted, is

$$b_1 = \frac{[(1+b_1^{\rm L})\sigma_8]}{\sigma_8(z_{\rm eff})}, \quad b_2 = \frac{[b_2^{\rm L}\sigma_8^2]}{\sigma_8^2(z_{\rm eff})} + \frac{8}{21}(b_1-1), \quad b_{s^2} = \frac{2\,[b_s^{\rm L}\sigma_8^2]}{\sigma_8^2(z_{\rm eff})} - \frac{4}{7}(b_1-1), \tag{10}$$

with the bracketed quantities the DESI chain values and σ_8 evaluated at the DESI fiducial cosmology. This note recommends against importing the DESI nuisance values at all, beyond the linear amplitude: applied to the draft's numbers, Eq. (10) gives $b_s^{\rm L} \simeq 2.5$–3.3 and $b_{s^2} \simeq +4$ to $+6$, which says that these particular chain values are dominated by the width of the DESI prior and by degeneracies with the counterterms, not by a measurement of the tidal field. A forecast fiducial should come from the calibrated relations of Section 2.3 (Table 1); the DESI posteriors then motivate the prior widths, not the central values.

5 Projection to thin-bin angular spectra

5.1 Separability and the D^4 rule

The exact (non-Limber) angular power spectrum between bins i and j is

$$C_\ell^{ij} = \frac{2}{\pi}\int dk\, k^2 \int d\chi_1\, W_i(\chi_1)\, j_\ell(k\chi_1) \int d\chi_2\, W_j(\chi_2)\, j_\ell(k\chi_2)\, \mathcal{P}(k; z_1, z_2), \tag{11}$$

where $W_i(\chi)$ is the bin's radial kernel (the normalized, Fingers-of-God-convolved $n(z)$ expressed in comoving distance χ), j_ℓ is the spherical Bessel function, and $\mathcal{P}$ is the unequal-time galaxy power spectrum. The computation is tractable because every term of the one-loop expansion is separable in k and time:

- the linear term carries $D(z_1)\,D(z_2)$;
- all five quadratic templates ($P_{\delta_1\delta_2}$, $P_{\delta_2\delta_2}$, $P_{\delta_1 s_2}$, $P_{\delta_2 s_2}$, $P_{s_2 s_2}$) are 22-type mode-coupling integrals over two linear spectra, so each carries $D^2(z_1)\,D^2(z_2)$;
- the 13-type pieces (the $\sigma_3^2 P_{\rm lin}$ term of $b_{3\rm nl}$ and the P_{13} inside the one-loop matter spectrum) carry the symmetrized $\frac{1}{2}\left[D^3(z_1)D(z_2) + D(z_1)D^3(z_2)\right]$.

At equal times every loop term scales as $D^4(z)$. This is exactly how both CCL and velocileptors evaluate the model: the loop templates are computed once from the $z = 0$ linear spectrum and rescaled by $D^4(z)$, with the bias coefficients supplying the remaining redshift dependence [8]. For the thin bins of this project the equal-time evaluation at the bin (or pair) mean redshift is accurate: across a $\Delta z = 0.02$ bin the growth factor changes by about 1%, the symmetric products above differ from $D^4(z_{\rm mid})$ only at second order in that 1%, and the residual is a few parts in 10^4 of the loop terms, which are themselves percent-level corrections.

The consequence for the FFTLog machinery is structural, and it is the one fact the implementation study needs from this section: adding the one-loop terms adds a fixed, small set of z-independent k-shapes (the six quadratic templates, $\sigma_3^2 P_{\rm lin}$, $k^2 P_{\rm lin}$, and a constant), each entering a bin's power spectrum with a scalar amplitude $b_X b_Y D^4(z_{\rm bin})$. The non-Limber cost scales with the number of distinct k-shapes, not with the number of bins. The project's low-rank engine projects the full per-bin $P_{gg}(k)$ through an SVD of its square root, so it carries the one-loop sum with no structural change; only the sign structure matters there, because the loop sum can turn P_{gg} slightly negative at low k for extreme bias values, and a square-root-based engine must guard against that.

5.2 Linear redshift-space distortions and counterterms for thin shells

Linear (Kaiser) redshift-space distortions enter Eq. (11) through the transformation of the density kernel: each factor of $f\mu^2$ in the flat-sky spectrum becomes a second derivative of the Bessel kernel in the exact projection,

$$\Delta_\ell^i(k) = \int d\chi\, W_i(\chi) \left[b_1(z)\, j_\ell(k\chi) - f(z)\, j_\ell''(k\chi) \right] D(z), \tag{12}$$

the form the project's covariance already implements. For wide photometric bins the j_ℓ'' term integrates down by orders of magnitude; for shells thinner than the scales probed it survives at full strength, because the shell resolves line-of-sight modes with μ near one. This is why thin-bin C_ℓ^{gg} recovers the Kaiser information of a three-dimensional analysis [11], and it cuts both ways: every μ-dependent term of Eq. (8) that a wide-bin analysis could drop is alive here.

The counterterms follow the same rule. The $\alpha_0 k^2 P_{\rm lin}$ piece is μ-independent and projects like the density: one extra k-shape, $k^2 P_{\rm lin}(k)$, with per-bin amplitude $\alpha_0 D^2(z_{\rm bin})$ (the counterterm multiplies the linear spectrum, so it scales as D^2, not D^4). The $\alpha_2 \mu^2 k^2 P_{\rm lin}$ piece projects through one j_ℓ'' in one of the two Bessel kernels, the same transfer structure as the Kaiser term; for thin shells its reach is comparable to α_0's rather than suppressed. The project's Fingers-of-God convolution of each bin's $n(z)$ with the velocity-dispersion kernel σ_v is a nonperturbative stand-in for part of this physics: it damps radial power with a μ-dependent, scale-dependent shape set by the sampled σ_v, and is therefore partially degenerate with α_2. Partial degeneracy is not identity: the convolution is a positive damping of fixed functional form, while α_2 can take either sign and scales as $k^2 P_{\rm lin}$.

5.3 Stochastic terms under projection

A constant three-dimensional power SN_0 projects to

$$C_\ell^{ij}\big|_{\rm stoch} = \mathrm{SN}_0 \int d\chi\, \frac{W_i(\chi)\, W_j(\chi)}{\chi^2}, \tag{13}$$

independent of ℓ. Equation (13) is exact, not a Limber approximation: for a constant $\mathcal{P}$ the k integral of Eq. (11) collapses by Bessel orthogonality, so the Limber formula is exact for white noise. For a top-hat bin of width $\Delta\chi$ the result is $\mathrm{SN}_0/(\chi^2\Delta\chi)$, and at the Poisson value $\mathrm{SN}_0 = 1/\bar{n}$ it reproduces the familiar angular shot noise $1/\bar{n}_{2D}$, with $\bar{n}_{2D}$ the surface density of the bin. The angular pipeline already carries the Poisson part in the covariance as the noise spectrum N_ℓ, so the free parameter in the mean model is the deviation from Poisson, not the full $1/\bar{n}$. Two projection-specific effects deserve attention. First, the Fingers-of-God convolution makes adjacent thin-bin kernels overlap, so Eq. (13) is nonzero for neighboring pairs: a stochastic term leaks into the cross spectra that the project measures, with a magnitude set by the kernel overlap integral. Second, the $\mathrm{SN}_2 k^2 \mu^2$ term projects with the radial-derivative structure of Section 5.2 and grows as k^2; Section 6 shows it stays below the other terms at this project's cuts.

5.4 What an angular analysis can drop

Terms that can be dropped at both of this project's cuts: the μ^4 members (α_4, SN_4), which DESI itself drops with the hexadecapole and whose thin-shell projections are suppressed relative

to their μ^2 partners; the $b_{3\mathrm{nl}}$ term, bounded by co-evolution at $|b_{3\mathrm{nl}}| \leq 0.17$ and worth at most 0.6% of P_{gg} at $k \leq 0.1\,h\,\mathrm{Mpc}^{-1}$ (Table 3), and degenerate with α_0 in shape over the fitted range; and the derivative bias b_{k^2}, exactly degenerate with α_0 at one loop. Terms that cannot be dropped at the shipped mask: the b_2 and b_{s^2} loops, α_0, a radial counterterm degree of freedom (an α_2 equivalent or a demonstrably sufficient σ_v freedom), and SN_0. The quantitative basis follows.

6 The size of every term

Table 3 gives the fractional contribution of each one-loop term to P_{gg}, computed with FAST-PT 4.0 and CAMB 1.6.5 in the group's `ccl` conda environment at the project fiducial cosmology (the `cocoapy311` environment carries neither package; Appendix C records the setup). The reference is the linear-bias model $b_1^2 P_{\mathrm{mm}}$ with halofit, the quantity the current likelihood computes. Biases follow Table 1; the counterterm column shows the one-sigma DESI prior $\alpha_0 = 12.5\,h^{-2}\mathrm{Mpc}^2$ as a size scale; the shot-noise column shows the Poisson power $1/\bar{n}$ from the project's own thin-bin densities, which the covariance carries as noise, as context for the stochastic freedom.

Reading the table at the project's two reference cuts:

At $k_{\mathrm{max}} = 0.075\,h\,\mathrm{Mpc}^{-1}$ (the non-Limber UV-fraction cut that matches the thin-bin radial resolution), the net loop correction is 1–3% of P_{gg}, with individual terms up to 3% and partial cancellations that depend on the bias values. The counterterm prior scale is 1–3%. A 30% non-Poisson stochastic residual, the size Nicola et al. [9] detect for red samples, is 3% of P_{gg} for the LRG chunks and more for ELG. Against the forecast's statistical power (sub-percent amplitude precision per chunk once all bins and bands are aggregated), none of these is negligible; the saving grace is that below $0.075\,h\,\mathrm{Mpc}^{-1}$ their scale dependence is mild, so the free per-chunk linear amplitude absorbs most of each term and the damage concentrates in the residual shape. That is an argument for a small extension, not for linear bias alone.

At the shipped mask, $k \simeq 0.18\,h\,\mathrm{Mpc}^{-1}$, the picture changes qualitatively. The $b_1 b_2$ term alone reaches 7–17% with strong scale dependence; the quadratic-quadratic terms reach 3–7%; the counterterm scale is 10–19%. These are not absorbable by an amplitude, and their shapes differ from one another, so the full one-loop parameter set is required, as the DESI DR1 choice of $k_{\mathrm{max}} = 0.2\,h\,\mathrm{Mpc}^{-1}$ with seven nuisances per tracer already reflects.

7 The draft model against DESI's

Point by point, the draft's Eulerian model against the DESI DR1 full-shape baseline it cites:

- **The six quadratic templates match CCL exactly**, and CCL matches the McDonald–Roy structure that FAST-PT implements. No issue beyond the `sub_lowk` statement of Section 2.1.
- **The bias values need the conversion of Section 4** before they can enter an Eulerian calculator, and this is the largest single correction. At the table's numbers the tidal term enters with the opposite sign and several times the magnitude of the co-evolution expectation (the BGS $b_1 b_{s^2}$ product is $+1.96$ against -0.50), and the quadratic coefficient differs at a comparable level, so the loop sum differs from the converted one by several percent of P_{gg} at $k \simeq 0.1\,h\,\mathrm{Mpc}^{-1}$, the size of the whole correction being modeled. The linear column, read as an Eulerian bias, is low by 30–60% for the LRG and ELG chunks; a

k [h/Mpc]	$b_1 b_2$	$\frac{1}{4}b_2^2$	$b_1 b_{s^2}$	$\frac{1}{2}b_2 b_{s^2}$	$\frac{1}{4}b_{s^2}^2$	$b_1 b_{3\mathrm{nl}}$	loop sum	$\alpha_0 k^2 P_{\mathrm{lin}}$	$1/\bar{n}$
$z = 0.3$, BGS-like: $b_1 = 1.56$, $b_2 = -0.64$, $b_{s^2} = -0.32$, $1/\bar{n} = 171$									
0.050	−0.8	+2.0	+0.1	+1.0	+0.2	+0.1	+2.6	1.3	0.7
0.075	−2.3	+2.7	+0.3	+1.1	+0.2	+0.2	+2.2	2.9	1.0
0.100	−4.9	+3.9	+0.5	+1.4	+0.2	+0.3	+1.4	5.0	1.6
0.150	−10.8	+5.7	+0.9	+1.5	+0.3	+0.4	−2.0	10.7	2.7
0.200	−17.2	+7.2	+1.3	+1.6	+0.3	+0.5	−6.3	17.7	4.0
$z = 0.5$, LRG-like: $b_1 = 2.20$, $b_2 = +0.28$, $b_{s^2} = -0.69$, $1/\bar{n} = 2463$									
0.050	+0.2	+0.2	+0.2	−0.4	+0.3	+0.1	+0.6	0.7	6.4
0.075	+0.6	+0.2	+0.4	−0.4	+0.3	+0.2	+1.3	1.5	9.2
0.100	+1.2	+0.3	+0.6	−0.5	+0.4	+0.3	+2.4	2.5	14.2
0.150	+2.7	+0.5	+1.2	−0.6	+0.5	+0.5	+4.7	5.4	24.2
0.200	+4.4	+0.6	+1.6	−0.6	+0.6	+0.7	+7.2	9.1	36.2
$z = 0.8$, LRG-like: $b_1 = 2.54$, $b_2 = +1.09$, $b_{s^2} = -0.88$, $1/\bar{n} = 2312$									
0.050	+0.5	+1.4	+0.1	−1.1	+0.3	+0.1	+1.3	0.5	6.0
0.075	+1.5	+1.9	+0.3	−1.2	+0.3	+0.2	+2.9	1.1	8.6
0.100	+3.1	+2.7	+0.5	−1.5	+0.4	+0.3	+5.5	1.9	13.3
0.150	+7.1	+4.0	+1.0	−1.7	+0.5	+0.4	+11.2	4.1	22.9
0.200	+11.5	+5.1	+1.4	−1.8	+0.5	+0.6	+17.4	7.0	34.6
$z = 1.3$, ELG-like: $b_1 = 1.56$, $b_2 = -0.64$, $b_{s^2} = -0.32$, $1/\bar{n} = 2913$									
0.050	−0.3	+0.8	+0.1	+0.4	+0.1	+0.0	+1.0	1.3	31.0
0.075	−0.9	+1.1	+0.1	+0.4	+0.1	+0.1	+0.9	2.9	43.9
0.100	−2.0	+1.6	+0.2	+0.5	+0.1	+0.1	+0.6	5.1	69.1
0.150	−4.5	+2.4	+0.4	+0.6	+0.1	+0.2	−0.8	11.1	119.8
0.200	−7.4	+3.2	+0.5	+0.7	+0.1	+0.2	−2.7	19.1	183.8

Table 3: Each one-loop term of Eq. (2) in percent of the linear-bias model $b_1^2 P_{\mathrm{mm}}$ (halofit), at the co-evolution biases of Table 1. Loop templates are FAST-PT at $z = 0$ scaled by $D^4(z)$, with the low-k constants kept (CCL's default, `sub_lowk` off). The counterterm column evaluates $\alpha_0 k^2 P_{\mathrm{lin}}(k, z)$ at the DESI one-sigma prior width $\alpha_0 = 12.5\, h^{-2}\mathrm{Mpc}^2$. The last column is the Poisson power from the project's DESI DR3-forecast thin-bin densities ($\bar{n} = 5.8 \times 10^{-3}$, 4.1×10^{-4}, 4.3×10^{-4}, 3.4×10^{-4} in $(h/\mathrm{Mpc})^3$ at $z = 0.3, 0.5, 0.8, 1.3$); it sits in the covariance, not the model, and is listed to set the scale of the stochastic freedom. $1/\bar{n}$ values in the subheadings are in $(h^{-1}\mathrm{Mpc})^3$.

free amplitude absorbs that in a fit, while a fixed-fiducial use (a data vector, a covariance, a Fisher derivative) carries it.

- $b_3^{\rm L} = 0$ **is safe**, and is DESI's own baseline. The co-evolution value it discards is $|b_{3\rm nl}| \leq 0.17$, worth at most 0.6% of P_{gg} below $0.1\,h\,{\rm Mpc}^{-1}$, degenerate with α_0; Maus et al. [6] validate the choice directly. The drop is safe at both cuts provided α_0 stays in the model.

- $\alpha_0 k^2 P_{\rm lin}$ **against** $b_{k^2} k^2 P_{\rm mm}$: the same physical term in two dressings. They differ by $P_{\rm lin}$ versus $P_{\rm mm}$ (a 10–20% relabeling of the coefficient below $0.2\,h\,{\rm Mpc}^{-1}$) and by DESI's convention of factoring $(b_1 + f\mu^2)\,b_1\,\alpha_0$ out of the tree-level structure. CosmoLike's concrete form is $2\,b_1\,b_K\,k^2\,P_{\rm mm}$ with b_K the per-bin derivative-bias slot, so matching the draft's term needs $\alpha_0 = 2\,b_1\,b_K$ at the level of the k^2 coefficient plus the $P_{\rm lin}$-versus-$P_{\rm mm}$ relabeling. Any of the three is fine; the prior must be stated in the convention used, and the DESI width translates to a 50% correction to the linear spectrum at the analysis $k_{\rm max}$.

- **No redshift-space loops**: the draft keeps the loop sum in real space and attaches linear Kaiser projection plus the Fingers-of-God $n(z)$ convolution. The dropped μ-dependent loop and counterterm pieces are the thin-shell analogue of what DESI's α_2 and the one-loop velocity terms carry, and Section 5.2 shows thin shells do not suppress them. At $k_{\rm max} = 0.075\,h\,{\rm Mpc}^{-1}$ they are percent-level and partially degenerate with the sampled σ_v; the residual after marginalizing σ_v has not been measured, and that measurement (a Fisher bias of a data vector with redshift-space loops against the real-space-loop model) belongs to the implementation study. At $0.18\,h\,{\rm Mpc}^{-1}$ the drop is not defensible without that measurement, and the model needs an explicit radial counterterm.

- **No stochastic terms**: the Poisson part sits correctly in the covariance, but the non-Poisson residual is detected at tens of percent of Poisson for red samples [9], which is percent-level in P_{gg} for the LRG and ELG chunks (Table 3). A forecast that fixes it to zero converts halo-exclusion physics into cosmological information. One $\rm SN_0$ per chunk is required at either cut; the `sub_lowk` constants fold into the same parameter.

- **One constant bias set per DESI chunk**: the draft fixes (b_1, b_2, b_{s^2}) per chunk while the pipeline evolves $b_1 = B_g/D(z)$ across the chunk's thin bins. The two statements disagree with each other, and the constant option disagrees with co-evolution: if b_1 drifts by 7–20% across a chunk, b_{s^2} drifts with it (Table 1, last column). Section 8 resolves this.

- **Halofit** $P_{\delta\delta}$: the hybrid is validated for projected statistics to beyond the shipped mask [9]. The requirements are consistency (the same $P_{\rm mm}$ in the data vector, covariance and Fisher derivatives) and awareness that halofit's accuracy, not the bias expansion, can set the error budget at the mask edge.

8 Bias evolution within a group

The project's correlated-prior note (`notes/correlated_bias_prior`) models the linear bias of bin i in group g as $b_i = B_g(1+\delta_i)/D(z_i)$, with the fractional deviations δ_i drawn from a Gaussian prior correlated across neighboring bins. The higher-order sector should follow the same logic. Co-evolution makes the trend deterministic: if $b_1(z) = B_g/D(z)$, then within the group

$$b_{s^2}(z) = -\frac{4}{7}\left(\frac{B_g}{D(z)} - 1\right) + \Delta_{s,g}, \qquad b_2(z) = b_2^{\rm fit}\left(\frac{B_g}{D(z)}\right) + \Delta_{2,g}, \tag{14}$$

with $b_2^{\rm fit}$ the Lazeyras relation of Eq. (7) and $\Delta_{2,g}$, $\Delta_{s,g}$ one free offset per group. The offsets absorb the known imperfection of the calibrated relations (the 30% spread between co-evolution and the excursion-set γ_2 at LRG biases, and the scatter of the Lazeyras fit); the deterministic part carries the within-group evolution, which is not small: across LRG3 the co-evolution b_{s^2} moves from -0.88 to -1.09, a drift comparable to the offset prior width itself. Holding b_2 and b_{s^2} constant while b_1 evolves as B_g/D is internally inconsistent; freeing them per bin (152 extra parameters for 76 bins) throws away the physical correlation the correlated-prior note was written to keep.

The parameter count under Eq. (14) is two offsets per group, twelve in total for the six chunks, and the correlated prior extends naturally: the offsets of chunks of the same tracer (LRG1–3, ELG1–2) share selection physics, so a Gaussian prior correlated across those chunks, with the same exponential kernel in redshift the linear-bias note uses, encodes "neighboring chunks deviate together" at second order. If the forecast later needs per-bin freedom in the higher-order sector, the same construction (per-bin offsets with the correlated kernel) bounds the cost without letting 76 independent b_2 values absorb the dark-energy signal.

9 Recommendation

At $k_{\rm max} = 0.075\,h\,{\rm Mpc}^{-1}$ (the cut matched to the thin-bin radial resolution), the minimal defensible model per DESI chunk, called model A below, is

- $b_1(z) = B_g/D(z)$ with B_g free, flat prior, as now;
- $b_2(z)$ from the Lazeyras relation on $b_1(z)$ plus a free offset $\Delta_{2,g}$, Gaussian prior of width 1.0 centered on zero;
- $b_{s^2}(z)$ from co-evolution on $b_1(z)$ plus a free offset $\Delta_{s,g}$, Gaussian prior of width 0.5 centered on zero (the width covers the co-evolution-versus-excursion-set spread at LRG biases);
- $b_{3{\rm nl}}$ fixed to co-evolution (or zero; the difference is inside the α_0 freedom);
- one counterterm α_0 per chunk, Gaussian prior centered on zero with the DESI rule rescaled to this cut: a 50% correction at $k_{\rm max} = 0.075\,h\,{\rm Mpc}^{-1}$ means a width of $0.5/k_{\rm max}^2 \simeq 89\,h^{-2}{\rm Mpc}^2$;
- one stochastic amplitude ${\rm SN}_0$ per chunk, Gaussian prior of width twice the Poisson power, following DESI;
- the sampled σ_v groups kept as the radial damping freedom, with the implementation study measuring whether they cover the dropped μ-dependent terms at this cut.

That is four new parameters per chunk (24 total) on top of the existing amplitudes and dispersions, against DESI DR1's seven per tracer at $k_{\rm max} = 0.2\,h\,{\rm Mpc}^{-1}$. The reasoning: at this cut every correction is 1–3% with mild scale dependence, so calibration-centered Gaussian freedom is enough, and the offsets keep the model honest about the calibrations without opening flat directions.

At $k_{\rm max} \simeq 0.15$–$0.2\,h\,{\rm Mpc}^{-1}$ (the shipped mask), the model, called model B below, must become the full one-loop set per chunk: b_2 and b_{s^2} free with the calibration relations demoted to prior centers (widths of order 2 and 1), α_0 at the DESI width $12.5\,h^{-2}{\rm Mpc}^2$, an explicit radial

counterterm (the $\alpha_2 \mu^2 k^2 P_{\rm lin}$ term projected through the j_ℓ'' transfer, or a validated σ_v-based equivalent), SN_0 and SN_2, and the redshift-space one-loop question settled by measurement rather than assumption. That is six to seven parameters per chunk, the DESI DR1 budget, and the honest statement is that the real-space-loop shortcut is unvalidated at this cut: until a Fisher-bias test shows otherwise, headline numbers from the shipped mask carry a modeling systematic of the size of the α_2-equivalent terms. Sampling in a physical basis ($b_1\sigma_8$-like combinations) is worth adopting at either cut, for DESI's reason: it tracks what the data constrain and reduces projection effects.

In both configurations, the fiducial higher-order values are the co-evolution and Lazeyras values of Table 1, not imports of DESI chain values, for the reason Section 4 ends with; and the covariance and Fisher machinery must adopt the same model as the data vector, including the stochastic sector.

10 Implementation in CoCoA

This section reports the implementation study that ran alongside this note: an inventory of what CoCoA/CosmoLike computes today for one-loop galaxy bias, the relation of the draft's CCL data vector to it, and the gaps against models A and B of Section 9. No code was changed. File references are to `external_modules/code/cosmolike_core/cosmolike` (the `lowrank_fftlog` branch) and to `projects/lssty1xdesi`.

10.1 What CosmoLike computes

The one-loop machinery of Section 2 largely exists in the C code; what is missing sits at the edges (the cross pairs, the stochastic sector) and in what the project likelihood passes.

- **Loop templates.** `get_FPT_bias` in `pt_cfastpt.c` computes the five quadratic templates $P_{\delta_1\delta_2}$, $P_{\delta_2\delta_2}$, $P_{\delta_1 s_2}$, $P_{\delta_2 s_2}$, $P_{s_2 s_2}$ from thirteen C-FAST-PT J terms at $z = 0$, rescaled by D^4 inside the projection integrands, exactly the separable structure of Section 5.1; the tables are cosmology-dependent and rebuilt when the cosmology changes. A second producer, `set_bias_PS` in `generic_interface.cpp`, can install a Python FAST-PT table instead. The $b_{3\rm nl}$ template $P_{\delta_1\psi}$ is the exception: `tab_d1d3` in `pt_cfastpt.h` is a fixed, cosmology-independent table, acceptable only while $b_{3\rm nl} = 0$.

- **Low-k constants.** The galaxy-clustering Limber integrand in `cosmo2D.c` subtracts the white-noise constants of Section 2.1 ($P_{\delta_2\delta_2} - 2\sigma_4$ and partners, with σ_4 read as half the low-k limit of $P_{\delta_2\delta_2}$): the `sub_lowk`-on convention, against CCL's default.

- **Bias slots and spectra.** `bias.c` carries per-bin parameters `gb1`, `gb2`, `gbs2`, `gb3`, `gbmag`, `gbK`. `C_gg_tomo_limber_work` assembles the auto integrand $b_1^2 P_{\rm mm} + 2\, b_1 b_K k^2 P_{\rm mm}$ plus D^4 times the six loop terms of Eq. (2); `C_gs_tomo_limber_work` and `C_gk_tomo_limber_work` carry the matching galaxy–lensing cross terms ($\frac{1}{2} b_2 P_{\delta_1\delta_2} + \frac{1}{2} b_{s^2} P_{\delta_1 s_2} + \frac{1}{2} b_{3\rm nl} P_{\delta_1\psi} + b_K k^2 P_{\rm mm}$).

- **Non-Limber engines.** `C_gg_lowrank_decomp_work`, the low-rank engine, builds its per-bin SVD of $\sqrt{P_g}$ from the complete one-loop Limber integrand, so the auto spectra are already exact in the one-loop P_g; it clamps $P_g \geq 10^{-15}\, b_1^2 P_\delta$ where the loop sum turns negative, the guard Section 5.1 called for. The classic FKEM core (`C_cl_tomo`) projects

only $b_1 P_{\rm lin}$ exactly and moves everything else through its Limber difference, so its one-loop content is Limber-only: the failure mode with scale-dependent bias at thin bins that the draft paper describes. The adjacent-bin cross pairs (`C_gg_cross_tomo_limber_work`) stop with `log_fatal` when second-order bias is active: not implemented.

- **Redshift-space distortions and stochastic terms.** Linear Kaiser projection (the j_ℓ'' transfer) plus the σ_v Fingers-of-God smearing of each $n(z)$; no one-loop redshift-space terms, no μ-dependent counterterms, and no SN_0 or SN_2 slots anywhere.

- **Interface and likelihood.** `set_nuisance_bias_fastpt` in `generic_interface.cpp` (arguments `B1`, `B2`, `B_MAG`, `B3nl`, `BK`), bound as `set_nuisance_bias` in the project's `interface/interface.cpp`, takes no tidal argument: `set_nuisance_nonlinear_bias` sets $b_{s^2} = -\frac{4}{7}(b_1 - 1)$ per bin whenever $b_2 \neq 0$, the co-evolution relation of Eq. (6) in the CCL normalization. The project likelihood (`likelihood/desi_cl.py`) passes linear bias only (`B2` = `B3nl` = `BK` = `B_MAG` = 0), so none of the machinery above is exercised by lssty1xdesi today.

- **Fisher and covariance.** The Fisher module, `cosmolike_lssty1xdesi_fisher.py` under `interface/`, varies every declared `DESI_` parameter, so new nuisances enter it automatically once the likelihood reads them; C_ℓ is linear in b_2, b_{s^2}, b_K and SN_0 at fixed b_1 apart from the quadratic $\frac{1}{4}b_2^2$ and $\frac{1}{4}b_{s^2}^2$ terms, so the new derivatives are cheap. The covariance (`covariances/spectra_cov.c`) carries linear bias only, on top of the open mismatch that it takes the raw chunk amplitudes instead of $B_g/D(z)$.

10.2 The draft's data vector against CosmoLike

The two calculations share the six templates in the same normalization (CCL, FAST-PT and CosmoLike agree term by term). They differ in three places. First, the low-k constants: CosmoLike subtracts them, and a CCL vector at the CCL default does not, so the two vectors differ by the white-noise term quantified in Section 2.1 until the draft states its `sub_lowk` setting. Second, the k^2 term: the draft adds $\alpha_0 k^2 P_{\rm lin}$ while CosmoLike has $2\,b_1 b_K k^2 P_{\rm mm}$; matching them needs $\alpha_0 = 2\,b_1 b_K$ at the level of the k^2 coefficient plus the $P_{\rm lin}$-versus-$P_{\rm mm}$ relabeling (Section 7). Third, the tidal bias: the interface fixes b_{s^2} to co-evolution, so CosmoLike cannot take the draft table's tidal values; given Section 4, the co-evolution default is closer to the physical value than those entries. The implementation study's independent check of that verdict is folded into Section 4.

10.3 Gaps against model A

Model A (Section 9: per chunk, the b_2 and b_{s^2} trends with group offsets, $b_{3\rm nl}$ from co-evolution or zero, one α_0, one SN_0, the σ_v groups kept) decomposes into the following work items.

need	status	change
$b_2(z)$ trend + offset per group	per-bin `gb2` exists in C	likelihood only: compute per-bin values from $b_1(z)$ and a new `DESI_B2OFF` parameter per group in `desi_cl.py` (small, Python)
b_{s^2} trend + offset	C slot exists; the interface forces co-evolution	add a tidal argument to `set_nuisance_bias`, keeping the co-evolution default when absent; likelihood computes the per-bin values (small, C++ interface and Python)
$b_{3\mathrm{nl}}$	fixed template table	keep $b_{3\mathrm{nl}} = 0$ (the DESI baseline); make `tab_d1d3` cosmology-dependent before ever freeing it
α_0 counterterm	$2\,b_1 b_K k^2 P_{\mathrm{mm}}$ exists	reuse b_K with the stated mapping, or add a $k^2 P_{\mathrm{lin}}$ variant for an exact match to the draft and DESI (small)
SN_0 per chunk	missing	new nuisance slot; the projection is the exact Eq. (13), autos plus the overlap of Fingers-of-God-convolved adjacent kernels (small, C)
non-Limber autos	low-rank engine already exact in the one-loop P_g	none
non-Limber adjacent cross pairs	`log_fatal` with second-order bias	the main C task: each template separates as $D^2(z_1)D^2(z_2)$ (Section 5.1), so one extra FFTLog projection per template, or an extension of the low-rank cross factors to the one-loop P_g, with the Limber subtraction rule unchanged
classic FKEM	one-loop content Limber-only	use the low-rank engine on thin bins; document that classic FKEM is not valid with one-loop bias there
covariance	linear bias only	add the one-loop P_{gg} together with the $B_g/D(z)$ bias fix to `spectra_cov.c` (medium)
Fisher	prefix-based	none beyond the likelihood parameters

10.4 Model B and the order of work

Model B adds free b_2 and b_{s^2} with calibration-centered priors (yaml work once model A is in), a radial counterterm projected with the j''_ℓ transfer like the Kaiser term, SN_2, and the redshift-space one-loop terms. CosmoLike has no redshift-space loop templates and FAST-PT's redshift-space terms are not in the C port, so the options are: absorb them in the sampled σ_v

plus an explicit α_2-like counterterm and measure the Fisher bias against a full redshift-space model, or port the FAST-PT redshift-space templates (or an external $P(k,\mu)$ emulator). Model B is a research task, not an extension of model A.

The recommended order:

1. interface: free the tidal argument (co-evolution default), add the SN_0 slot, state the b_K-to-α_0 mapping;
2. likelihood: group-level trends and offsets for b_2 and b_{s^2}, α_0, SN_0, with the priors of Section 9;
3. C: one-loop for the adjacent cross pairs through the separable $D^2(z_1)D^2(z_2)$ templates;
4. validation: CosmoLike's Limber one-loop against CCL's `EulerianPTCalculator` with the same templates and the same `sub_lowk` choice (the `ccl` conda environment carries pyccl and FAST-PT; `cocoapy311` carries neither), then the low-rank non-Limber against a brute-force Bessel integral for one thin pair;
5. covariance: the one-loop P_{gg} and the $B_g/D(z)$ bias fix together;
6. model B only after a measured Fisher bias of the redshift-space loops.

A Bases and conversions, with every factor

A.1 The operators

The tidal field of a density field δ is $s_{ij} = (\partial_i\partial_j/\partial^2 - \delta_{ij}/3)\delta$, with $s^2 = s_{ij}s_{ij}$. Writing the normalized potential $\phi = \partial^{-2}\delta$, expanding the square gives $s^2 = (\partial_i\partial_j\phi)^2 - \frac{2}{3}\,\delta\,\partial^2\phi + \frac{1}{9}\,\delta_{ij}\delta_{ij}\,\delta^2 = (\partial_i\partial_j\phi)^2 - \frac{1}{3}\,\delta^2$, using $\partial^2\phi = \delta$ and $\delta_{ij}\delta_{ij} = 3$. The Galileon operator is $\mathcal{G}_2 = (\partial_i\partial_j\phi)^2 - \delta^2$, so

$$\mathcal{G}_2 = s^2 - \frac{2}{3}\,\delta^2. \tag{15}$$

A term $b_{\mathcal{G}_2}\mathcal{G}_2$ in δ_g therefore equals $b_{\mathcal{G}_2}s^2 - \frac{2}{3}b_{\mathcal{G}_2}\delta^2$: matching to Eq. (1), $b_{s^2} = 2\,b_{\mathcal{G}_2}$, and the quadratic coefficients of the two bases relate as $b_2 = b_2' - \frac{4}{3}\,b_{\mathcal{G}_2}$, where b_2' is the coefficient of $\delta^2/2$ in the $(b_2', b_{\mathcal{G}_2})$ basis. The third-order operator of Eq. (1) is $\psi = \eta - \frac{2}{7}s^2 + \frac{4}{21}\delta^2$ with $\eta = \theta - \delta$ the velocity-divergence difference [1, 5]; at one loop only one combination of the cubic operators survives renormalization, with the filter $\sigma_3^2(k)$ of [2] as its power-spectrum template.

A.2 Normalization bookkeeping

Three conventions circulate for the quadratic and tidal terms, and every difference discussed in Section 4 is a crossing between two of them:

convention	quadratic term	tidal term
CCL / FAST-PT / [2]	$(b_2/2)\,\delta^2$	$(b_{s^2}/2)\,s^2$
velocileptors Eulerian [5]	$(c_2/2)\,\delta^2$	$c_s\,s^2$
velocileptors Lagrangian [5, 6]	$(b_2^{\mathrm{L}}/2)\,\delta_0^2$	$b_s^{\mathrm{L}}\,s_0^2$

So $b_2 = c_2$ but $b_{s^2} = 2c_s$: the quadratic parameters agree across the Eulerian conventions while the tidal ones differ by the factor of two that Eq. (5) carries.

A.3 The rotation, from the co-evolution solution

The 8/21 and $-2/7$ of Eq. (4) follow from solving the coupled continuity and Euler equations for a tracer that starts with local Lagrangian bias and is then advected (the derivation followed here is [2]; the same coefficients appear as the bias-parameter rotation in [5]). At linear order the Lagrangian bias dilutes as $b_1^{\rm L}(z) \propto 1/D(z)$, equivalently $b_1(z)-1 \propto 1/D(z)$: the co-evolution statement and the project's constant-clustering-amplitude shape $B_g/D(z)$ coincide for the linear bias up to the additive 1. At second order the evolved overdensity picks up, next to the gravitational F_2 kernel, the terms

$$\delta_g^{(2)} \supset \left[\frac{b_2^{\rm L}}{2} + \frac{4}{21} b_1^{\rm L}\right] \delta^2 \; - \; \frac{2}{7}\, b_1^{\rm L}\, s^2, \tag{16}$$

which, matched to $(c_2/2)\delta^2 + c_s s^2$, give $c_2 = b_2^{\rm L} + \frac{8}{21} b_1^{\rm L}$ and $c_s = b_s^{\rm L} - \frac{2}{7} b_1^{\rm L}$ once an initial $b_s^{\rm L} s_0^2$ is carried along. At third order the same procedure yields the dynamically generated piece $b_{3\rm nl} = \frac{32}{315}\, b_1^{\rm L}$ of Eq. (6) [2]. In the CCL normalization the tidal entries double, as Eq. (5) records.

A.4 Worked example: LRG1

DESI samples $(1 + b_1^{\rm L})\sigma_8(z_{\rm eff})$, $b_2^{\rm L}\sigma_8^2$, $b_s^{\rm L}\sigma_8^2$ at $z_{\rm eff} = 0.510$. Suppose the chain value of the first is 1.448 and $\sigma_8(0.510) = 0.649$ at the fiducial cosmology. Then $b_1 = 1.448/0.649 = 2.23$, and the co-evolution fiducials for the forecast are $b_{s^2} = -\frac{4}{7}(1.23) = -0.70$ and $b_{3\rm nl} = \frac{32}{315}(1.23) = +0.125$, with $b_2(b_1{=}2.23) = +0.34$ from Eq. (7). Using 1.448 itself as an Eulerian bias, as the draft table does, is a 31% error on b_1 and a factor b_1^2 error of 69% on the leading term of P_{gg}, before the tidal-sign error enters.

B The loop templates and their low-k constants

With F_2 and $S_2 = \mu_{12}^2 - \frac{1}{3}$ the standard second-order density and tidal kernels (μ_{12} the cosine between the two internal wavevectors), the quadratic templates of Eq. (2) are

$$\begin{aligned}
P_{\delta_1\delta_2}(k) &= 2\int_{\mathbf{q}} F_2(\mathbf{q}, \mathbf{k}-\mathbf{q})\, P_{\rm lin}(q)\, P_{\rm lin}(|\mathbf{k}-\mathbf{q}|), & P_{\delta_2\delta_2}(k) &= 2\int_{\mathbf{q}} P_{\rm lin}\, P_{\rm lin}, \\
P_{\delta_1 s_2}(k) &= 2\int_{\mathbf{q}} F_2\, S_2\, P_{\rm lin}\, P_{\rm lin}, & P_{\delta_2 s_2}(k) &= 2\int_{\mathbf{q}} S_2\, P_{\rm lin}\, P_{\rm lin}, \\
P_{s_2 s_2}(k) &= 2\int_{\mathbf{q}} S_2^2\, P_{\rm lin}\, P_{\rm lin}, & &
\end{aligned} \tag{17}$$

(arguments as in the first line throughout), evaluated by FAST-PT as FFTLog convolutions. The $k \to 0$ limits are $P_{\delta_2\delta_2} \to 2\sigma_4$, $P_{\delta_2 s_2} \to \frac{4}{3}\sigma_4$, $P_{s_2 s_2} \to \frac{8}{9}\sigma_4$ with $\sigma_4 = \int_{\mathbf{q}} P_{\rm lin}^2(q)$, white-noise constants that renormalize into the stochastic sector; CCL's `sub_lowk` subtracts exactly these three constants in the combinations shown in Eq. (2). The third-order template is $\sigma_3^2(k) P_{\rm lin}(k)$ with the filter σ_3^2 defined through the I_R kernel of [2]; its co-evolution-weighted contribution is the $b_1 b_{3\rm nl}$ column of Table 3. The unequal-time factors quoted in Section 5.1 follow from the perturbative order of each side of the correlator: 22-type products of two second-order fields carry $D^2(z_1)D^2(z_2)$, 13-type products of a third-order and a linear field carry the symmetrized $D^3 D$.

C Numerical setup for the size tables

Cosmology: the project fiducial, $\Omega_m = 0.3$, $\Omega_b = 0.04$, $h = 0.6732$, $10^9 A_s = 2.1$, $n_s = 0.966$, massless neutrinos, giving $\sigma_8(0) = 0.843$. Linear and halofit spectra from CAMB 1.6.5; loop templates from FAST-PT 4.0 on a 1200-point logarithmic grid with low- and high-k extrapolation and a FFTLog window of 0.75, mirroring CCL's `EulerianPTCalculator` defaults; growth factors from the evolution of the $k = 0.05\,\mathrm{Mpc}^{-1}$ mode. The packages come from the group's `ccl` conda environment (pyccl 3.3.3 is present there for cross-checks; the `cocoapy311` environment has neither fastpt nor pyccl). Thin-bin densities for the Poisson column: the project files `data/lssty1xdesi_desi_dr3_ngal.txt` (surface density per bin) and `..._bins.txt` (bin mean redshifts), converted to three-dimensional densities with $\bar{n} = n_{2D}\,/(\chi^2\,\Delta\chi)$ at $\Delta\chi = 62.28\,\mathrm{Mpc}$. The $\sigma_8(z_{\mathrm{eff}})$ values of Table 2 use the same cosmology; the DESI fiducial differs from it by a few percent in σ_8, which moves the implied biases coherently by the same few percent and does not affect the identification.

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